EE6405: Natural Language Processing (course coordinator: Dr S Supraja)

Course Aims

This course is intended to introduce to you the concepts of Natural Language Processing (NLP) from the fundamentals till the latest technologies and enable you to apply NLP techniques in real-world projects. It will introduce several major popular state-of-the-art NLP techniques with a focus on attention-based deep learning models. You would need to have exposure to programming skills (Python). An important takeaway would be the translation of theoretical concepts learnt into practical hands-on applications by running simulations on real-life datasets such as machine translation or text summarization. With the emergence of AI-based generative tools such as ChatGPT, NLP is becoming a highly valued topic in several research and development laboratories/sectors, hence this course will equip you with useful industry-relevant skills. Students taking this course will be equipped with the fundamental skillsets for machine learning and deep learning algorithms that can be applied in various contexts beyond NLP.

Intended Learning Outcomes (ILO)

By the end of this course, you should be able to:

1. Identify the appropriateness of NLP preprocessing techniques in different contexts
2. Explain the mathematical derivations of traditional NLP methods such as term weighting schemes, feature extraction techniques, and topic modeling
3. Execute simple NLP tasks such as classification and clustering on small-scale problems and evaluate the algorithm performance
4. Implement NLP algorithms in Python programming language
5. Distinguish between the theoretical concepts of traditional and deep neural network-based NLP techniques
6. Formulate the construction of word embeddings and model training using deep neural network-based architecture such as RNN, Seq2Seq, Attention mechanism, and transformers.
7. Describe the working principles of pre-trained language models
8. Design NLP-based project as a team to solve a real-life application by employing any techniques learnt along with fine-tuning the model

Course Content

Traditional

1. Bag-of-words
2. Preprocessing
3. Term weighting scheme
4. Feature extraction
5. Topic modeling
6. ML classifiers and clustering methods
7. Evaluation Metrics
8. Word Embeddings

Deep Neural Networks

9. Seq2Seq
10. Attention mechanism
11. Transformers and self-attention
12. Pretrained Language Models

13. Fine-tuning (hyper-parameter tuning)
14. Applications (chatbot, machine translation, sentiment analysis, summarisation, classification, generation)
15. Generative AI: Large Language Models
16. Emerging Trends and Industry Applications

Assessment (includes both continuous and summative assessment)

Note: It is advised that Group component and class participation should not be more than 40% and 20% respectively, unless with good justification.

Component	ILO Tested	Weighting	Team/Individual	Assessment Rubrics
Theory Quiz IRA/TRA	1,5,6,7	20%	Individual readiness assessment (15%) Team readiness assessment (5%)	*Appendix 1*
Mid-term Quiz	2,6,7	35%	Individual	*N.A.*
Coding Quiz	3,4	20%	Individual	*Appendix 2*
Final Capstone Group Video Project with group reflection	4,8	25%	Team	*Appendix 3*
Peer Evaluation	8* (not graded)	0%	Team	*Appendix 4*
Total		100%	Individual (70%) Team (30%)	

Description of Assessment Components:
There is no final examination for this course.

Theory quiz IRA/TRA: You are required to watch pre-recorded videos to understand the concepts taught at every stage of the course and answer a closed-book multiple-choice online quiz.

Mid-term quiz: This is a closed-book quiz taken online that tests the topics pertaining to traditional NLP, with a mix of short-structured and other question types.

Coding quiz: You are required to understand the concepts taught at every stage of the course. This is like a lab test, such that you would need to perform the analysis of Python codes and evaluation on the provided dataset.

Final Capstone group video project: In your team of 5-6 students, you will need to select an NLP task as your project topic. You are to design and implement a model to perform the analysis and evaluation on any dataset of your choice for the selected topic. You will submit a final video and present your work as a team with the thought process for each design step e.g., framework, hyperparameter, evaluation metric, etc. You would also need to submit a group reflection highlighting the contributions of every member and acknowledged by every other member in the team.

Peer evaluation: This is not graded, but compulsory. The purpose is to evaluate every team member for his/her contribution towards the group project. Failure to do the peer evaluation will result in a grade deduction of 5% penalty.

Formative feedback

For the IRA/TRA and Coding quiz, automatic feedback will be provided via the online test platform if you had an incorrect attempt.

Learning and Teaching Approach

Approach	How does this approach support you in achieving the learning outcomes?
Project work	This will provide an opportunity for you to learn from one another and to become active participants in their learning. Coming together and creating a solution will help to increase the effectiveness in student learning. Such group-based project work helps students to develop skills valued by employers (such as problem solving, negotiation, conflict resolution, leadership, critical thinking and time management). In addition, the TRA also involves collaborative learning as members in a group will discuss and come to a consensus to choose the answer(s). This would encourage exchange of ideas and for students to better understand their teammates' viewpoints.
Blended learning/flipped classroom with active use of multi-media resources (TEL)	This course is a hybrid of online lectures and face-to-face tutorials. The pre-recorded lectures are for students to watch before attempting the IRA/TRA in-class. There is also a component of active and hands-on learning in-class with live coding exercises. This will permit flexibility of access to learning materials, activities and assessments and can help you to develop independent learning and critical thinking skills.
Student presentations	You will present your research and findings to the whole class. This supports peer-learning as well as developing research and presentation skills.
Industry partnership	It is important to collaborate with industry partners to achieve hands-on training with their user-friendly tools to perform NLP algorithms.

Reading and References (can indicate the years of publications for the references below?)

(1) Daniel Jurafsky, James H Martin. Speech and Language processing. Third Edition. 2000

(2) Ian Goodfellow and Yoshua Bengio and Aaron Courville. Deep Learning. An MIT Press book. 2015

(3) Jacob Eisenstein. Introduction to Natural Language Processing. 2019

Course Policies and Student Responsibilities

(1) General

You are expected to complete all assigned pre-class readings and activities, attend all seminar classes punctually and take all scheduled assignments and tests by due dates. You are expected to take responsibility to follow up with course notes, assignments and course related announcements for seminar sessions they have missed. You are expected to participate in all seminar discussions and activities.

(2) Absenteeism

Absence from class without a valid reason will affect your overall course grade. Valid reasons include falling sick supported by a medical certificate and participation in NTU's approved activities supported by an excuse letter from the relevant bodies.

If you miss a lecture, you must inform the course instructor via email prior to the start of the class.

Academic Integrity

Good academic work depends on honesty and ethical behaviour. The quality of your work as a student relies on adhering to the principles of academic integrity and to the NTU Honour Code, a set of values shared by the whole university community. Truth, Trust and Justice are at the core of NTU's shared values.

As a student, it is important that you recognize your responsibilities in understanding and applying the principles of academic integrity in all the work you do at NTU. Not knowing what is involved in maintaining academic integrity does not excuse academic dishonesty. You need to actively equip yourself with strategies to avoid all forms of academic dishonesty, including plagiarism, academic fraud, collusion and cheating. If you are uncertain of the definitions of any of these terms, you should go to the Academic Integrity Handbook for more information. Consult your instructor(s) if you need any clarification about the requirements of academic integrity in the course.

Course Instructors

Instructor	Office Location	Phone	Email
S Supraja (coordinator)	S1-B1B-57	-	supraja.s@ntu.edu.sg

The contact info of other instructor(s) (to be decided) will be provided to students at the beginning of each semester.

Appendix 1: Theory quiz (IRA and TRA)

Pre-lesson learning is a prerequisite for participation in team-based learning activities during seminars. It will be measured through individual and team quizzes, and accounts for 20% of your grade. To do well on this assessment you need to prepare for tutorials by reading the lecture notes beforehand. Your readiness will be assessed via a test in each session in the form of Individual Readiness Assessment (IRA) and Team Readiness Assessment (TRA). These will consist of 5 multiple-choice questions each. The questions will cover the material from the week's content and are designed to help you practice applying your knowledge and provide feedback on any gaps in your knowledge. There will be subtraction of points for incorrectly chosen answer option(s). (Every question is graded 10 points. There will be 5 options with multiple correct answers). Every student will get a randomized order of questions and answer options for the IRA while the entire class will get the same set of questions in the same order for the TRA (that involves discussion with fellow classmates). Only one attempt is possible. Note: The IRA and TRA are administered in a closed-book format and must be attempted in person. Students who are late or absent without any valid excuse are not allowed to take make-ups. Any attempt to do the quizzes while not in class or any attempt to help others do so is tantamount to cheating and will be dealt with severely. Any attempt to refer to course materials or external references during the quiz will be dealt with similarly. Respondus lockdown browser will be used.

Appendix 2: Coding quiz (Respondus lockdown browser)

Accounting for a total of 20% of your grade, every coding quiz will have 5 multiple-choice questions. There will be a single correct answer per question. This is done individually similar to the IRA. There will be zero marks if you have chosen near-miss answer(s). There will be negative marks for completely incorrect answers (such that a negative mark below zero can be obtained for a 10 points question). However, overall the marks will then be scaled to a range of 0-100.

Appendix 3: Final Capstone Group Project

AI policy: Students are required to sign the declaration on how they have used AI especially for the generation of code – if they had chosen "generated purely by AI tools," will be penalized.

The overall marks in the rubrics will be scaled down to 25%.

Criteria	Exemplary (4)	Proficient (3)	Average (2)	Poor (1)
Requirements (e.g., acc of x%; runtime < x secs; explore at least 2 algorithms)	Meet all project requirements and specifications	Meet most project requirements and specifications	Meet some project requirements and specifications	Does not meet the project requirements and specifications
Choice of application and use of dataset(s)	Different application with multiple datasets	Different application with single dataset	Standard application with multiple datasets	Standard application with single dataset
Framework	Great choice of model with extra effort to improve performance	Suitable choice of model for the task	Model works for the task but there are many better options	Model does not fit the task well
Experiment Design	Use of multiple relevant evaluation metrics and compare with strong baselines	Use of multiple relevant evaluation metrics, but lacks comparison with baselines	Use some suitable evaluation metrics	Wrong choice of evaluation metrics
Interpretation of model	Explains in detail beyond the chosen application on how the model could be extrapolated along with the physical meaning.	Explains the context in detail but interpretation superficially.	Explains the context in detail but does not explain their interpretation beyond the current context.	Briefly explains the context and physical meaning of the model
Readability	The code is easy to read, well organized and documented	The code is fairly easy to read, fairly well organized and documented	The code is readable but no documentation provided	The code is poorly written and hard to comprehend

Efficiency	The code runs efficiently without losing readability	The code runs fairly efficiently without losing readability	The code uses brute-force	The code seems to be copied from various places and does not gel well together (redundancies)
Reusability	Multiple parts of the code could be reused	Some parts of the code could be reused	Only a small part of the code could be reused	The code is not written for reusability (hard coding everywhere)
Presentation	Impressive demonstration and intuitive frontend interface	Provides demonstration and interface	Presents well but lacks demonstration/ interface	Presenter does not seem to know the topic well
Video skills	All members have participated and there is a good flow.	Some members are involved and there is a good flow.	Some members are involved, but there is no flow in the video.	Only one member is present in the video.

Appendix 4: Peer evaluation

1. All team members are expected to complete the teamwork evaluation for all other members in the team (i.e., self-assessment is not required); see evaluation criteria and rating descriptors on the following pages. Should a student fail to complete the teamwork evaluation of all other members in his/her group, 5% will be deducted from the student's final team assignment mark.

2. All assessments and qualitative comments are *confidential*.

3. Each member's average rating given by his/her team members will be used to determine the final team assignment marks awarded to each member as follows:

Average Rating Range (min: 1; max: 9)	**Marks Deduction (out of 100)**	**Final Adjusted Marks (out of 100 after scaling)**
≥ 6.5	0	Original team assignment mark (**M**)
≥ 5.5 to < 6.5	-5	**M** – 5
≥ 4.0 to < 5.5	-10	**M** – 10
≥ 3.0 to < 4.0	-15	**M** – 15
> 1.0 to < 3.0	-20	**M** – 20
= 1.0	-**M**	0

4. Please complete the teamwork evaluation through NTULearn Peerceptiv assignment platform for **ALL** your team members for the final capstone project.

5. If you feel you have reason to be concerned about the scores that you will receive from other members in your team, **please consult your instructor**.

6. Please note that the teaching team reserves the right to adjust students' final team assignment marks based on additional considerations including gathered information, certified special education needs (SEN), and medical diagnosis.

Teamwork Evaluation Criteria

Criteria (Weighs)	**Score from 1 to 9**[2,3] (**1**: Never; **3**: Rarely; **5**: Occasionally; **7**: Frequently; **9**: Always)				
(For 6-members team)	*Member A*	*Member B*	*Member C*	*Member D*	*Member E*
Member name					
a. Fulfilling one's responsibilities duly (15%)					
Behaved responsibly--such as attend meetings punctually and regularly; participate in discussion; complete assigned tasks/roles punctually.	*Score from 1 to 9*				
	Qualitative comments/reasons				
b. Fulfilling one's responsibilities effectively (25%)					
Behaved and contributed effectively--such as quality of work produced; creativity of ideas; extensiveness of research and thinking.	*Score from 1 to 9*				
	Qualitative comments/reasons				
c. Managing interpersonal relationships (30%)					
Listened attentively to and sought inputs from others; helped team resolve conflicts and achieved common understanding to function effectively; promoted respect for others and differences; fostered camaraderie.	*Score from 1 to 9*				
	Qualitative comments/reasons				
d. Providing support to others to achieve goals (30%)					
Behaved fairly and ethically—such as sharing responsibilities and giving credits. Exhibited group citizenship behavior--such as helping others to learn and complete their work through guidance and encouragement; standing up for others when needed.	*Score from 1 to 9*				
	Qualitative comments/reasons				

[2] Score of 1 should be given only when a team member does not really deserve to be awarded any mark for the team assignment (i.e., zero mark) because the member either has not or has barely participated and/or contributed to the team assignment in any meaningful manner.

[3] See detailed score descriptions for each criterion in the next table.

Teamwork Evaluation Criteria – Score Descriptions

a. Fulfilling one's responsibilities duly (15%)									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
	Never		**Rarely**		**Occasionally**		**Frequently**		**Always**
Behaved responsibly--such as attend meetings punctually and regularly; participate in discussion; complete assigned tasks/roles punctually.	Did not participate actively in team meetings at all (even though he/she might still be present.) Did not contribute any useful ideas that were relevant for getting the project completed. Did not submit any or usable work on time and never did so without prompting.		Rarely participated actively in team meetings. Rarely contributed useful ideas that were relevant for getting the project completed. Rarely submitted work on time and did so rarely without prompting.		Occasionally participated actively in team meetings. Occasionally contributed useful ideas that were relevant for getting the project completed. Occasionally submitted work on time and did so occasionally without prompting.		Frequently participated actively in team meetings. Often contributed useful ideas that were relevant for getting the project completed. Frequently submitted assigned work on time without prompting.		Always participated actively in team meetings. Always contributed useful ideas that were relevant for getting the project completed. Always submitted assigned work on time without prompting.
b. Fulfilling one's responsibilities effectively (25%)									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
	Never		**Rarely**		**Occasionally**		**Frequently**		**Always**
Behaved and contributed effectively--such as quality of work produced; creativity of ideas; extensiveness of research and thinking.	Quality could not be evaluated because nothing relevant or usable was submitted.		Quality of work could be improved vastly; required major revisions before submission. Work reflected superficial effort in completing the project.		Quality of work could be improved; required some revisions before submission. Work reflected some effort in completing the project.		Quality of work was of reasonably acceptable standards; required minor revisions before submission. Work reflected reasonable effort in completing the project.		Quality of work was of highly acceptable standards; required no revision before submission. Work reflected extensive effort in completing the project.

c. Managing interpersonal relationships (30%)									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
	Never		**Rarely**		**Occasionally**		**Frequently**		**Always**
Listened attentively to and sought inputs from others; helped team resolve conflicts and	Never listened to others. Never attempted to help team resolve conflicts. Never		Rarely listened to others. Rarely attempted to help team resolve conflicts. Rarely		Occasionally listened to others, with occasional attempts to help team resolve conflicts.		Frequently listened to others, with frequent attempts to help team resolve conflicts.		Always listened to others, always attempted to help team resolve conflicts.

achieved common understanding to function effectively; promoted respect for others and differences; fostered camaraderie.	attempted to achieve common understanding to function effectively as a team. Never promoted respect for others, with no attempt to foster camaraderie.		attempted to achieve common understanding to function effectively as a team. Rarely promoted respect for others, with rare attempts to foster camaraderie.		Occasionally attempted to achieve common understanding to function effectively as a team. Occasionally promoted respect for others, with occasional attempts to foster camaraderie.		Frequently attempted to achieve common understanding to function effectively as a team. Frequently promoted respect for others, with notable attempts to foster camaraderie.		Always attempted to achieve common understanding to function effectively as a team. Always promoted respect for others and demonstrated consistent attempts to foster camaraderie.
d. Providing support to others to achieve goals (30%)									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
	Never		**Rarely**		**Occasionally**		**Frequently**		**Always**
Behaved fairly and ethically—such as sharing responsibilities, and giving credits. Exhibited group citizenship behavior--such as helping others to learn and complete their work through guidance and encouragement; standing up for others when needed.	Never took on a fair share of responsibilities; never gave appropriate credits to others. Never attempted to help others learn; never provided guidance and encouragement. Never stood up for others when needed.		Rarely took on a fair share of responsibilities; rarely gave appropriate credits to others. Rarely attempted to help others learn; rarely provided guidance and encouragement. Rarely stood up for others when needed.		Occasionally took on a fair share of responsibilities; occasionally gave appropriate credits to others. Occasionally attempted to help others learn; occasionally provided guidance and encouragement. Occasionally stood up for others when needed.		Frequently took on a fair share of responsibilities; frequently gave appropriate credits to others. Frequently attempted to help others learn; frequently provided guidance and encouragement. Frequently stood up for others when needed.		Always took on a fair share of responsibilities; always gave appropriate credits to others. Always attempted to help others learn; always provided guidance and encouragement. Always stood up for others when needed.